

WASP-52b

R-0431

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_171207 · created 2026-07-19T10:55:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458094.6376 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.204 +/- 0.062
Transit depth (Rp/Rs)^2	4.17 +/- 2.53 [%]
Orbital Inclination (inc)	89.48 +/- 2.75
Airmass coefficient 1 (a1)	18.04 +/- 13.52
Airmass coefficient 2 (a2)	-2.21 +/- 0.95
Scatter in the residuals of the lightcurve fit is	105.23 %
Best Comparison Star	#3 - [380, 443]
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.0869 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.204 ± 0.062 vs 0.1646 (deviation 0.64σ)
Duration fit vs published	0.0869 d vs 0.0758 d (deviation 1.32σ)
Mid-time O–C	-9.1 min
Depth SNR	3.3
Residual scatter	105.23 %

Light curve

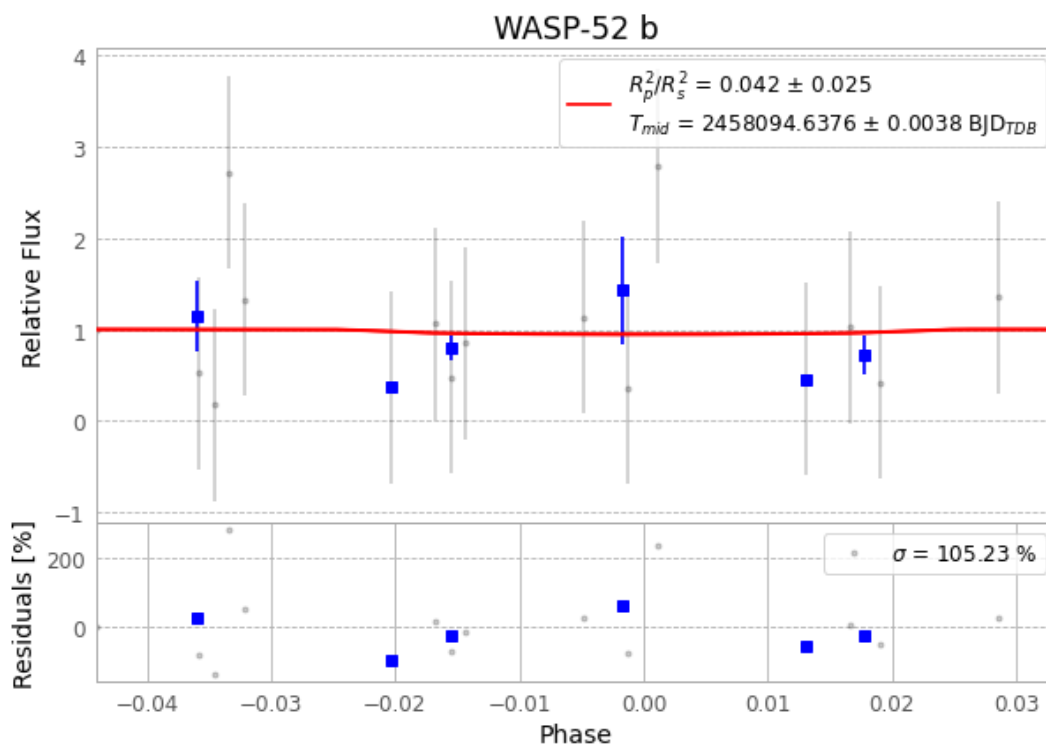


Figure 1 — FinalLightCurve_WASP-52 b_2017-12-06.png (SHA-256 6125eee4311953c2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 49], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4041f5c0753f4bde...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

84 FITS frames (55 MB), WASP-52171207011815.FITS → WASP-52171207053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171207.log (faa7cf644a1d...), FinalLightCurve_WASP-52 b_2017-12-06.png (6125eee43119...), FinalLightCurve_WASP-52 b_2017-12-06.csv (7f59767d7adf...)

Compute resources

Wall time 240.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 10:55Z.